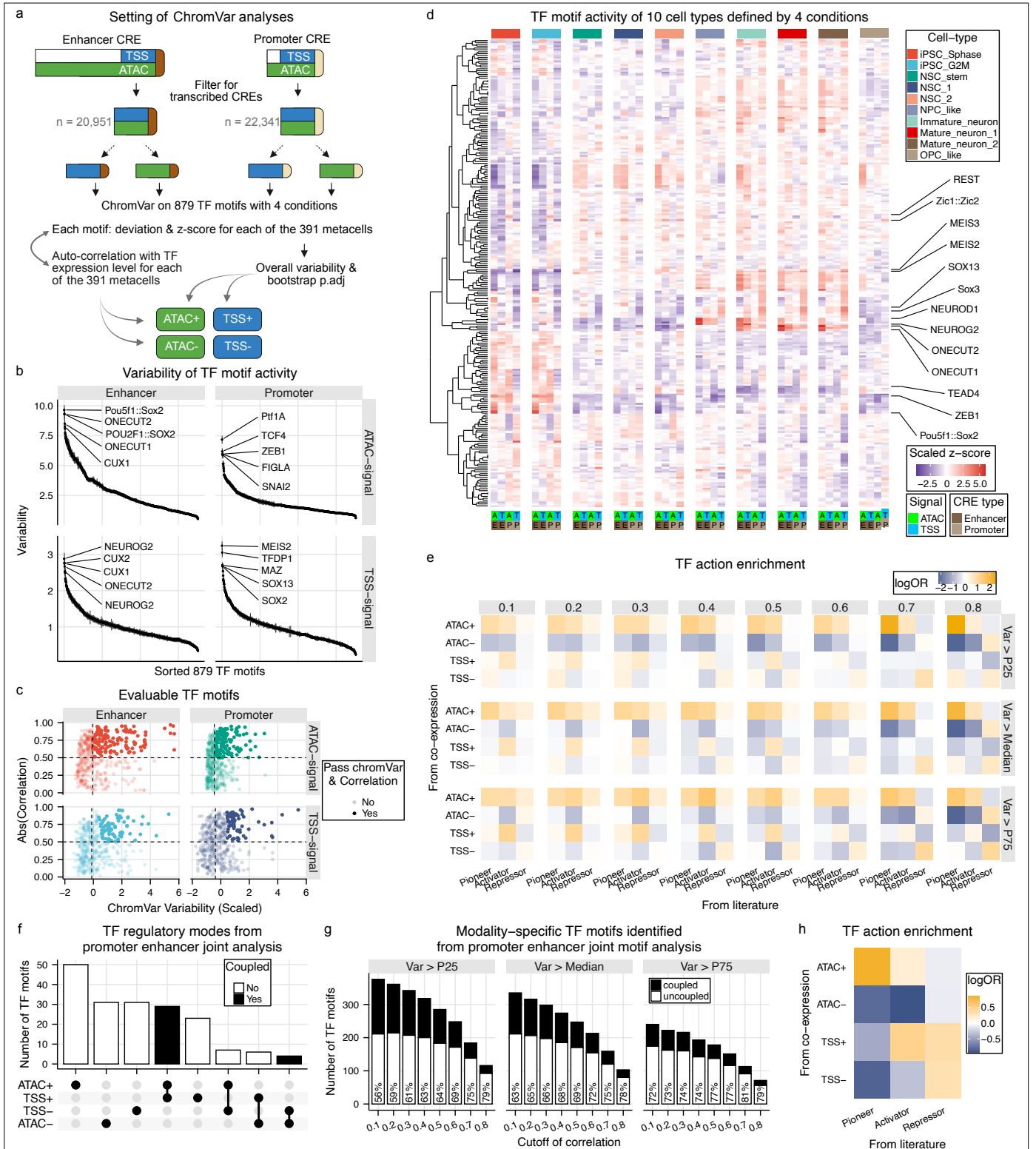




Extended Figure 3 | TF motif activity profiling from ChromVar and cross-modality correlation.

a, Workflow schematic for ChromVar analysis across four regulatory contexts. **b**, Variability of motif activity across metacells, comparing results from assessing enhancer CREs or promoter CREs using either the ATAC or TSS signals. Variability scores ($n = 879$ motifs) are shown with bootstrap confidence intervals. Top 5 most variable TF motifs are highlighted. **c**, Scaled ChromVar variability score and correlation coefficients for the 400 TF motifs with expression support across the four regulatory contexts. Motifs passing the thresholds are indicated in solid colors. **d**, Heatmap showing summarized motif activity across the 10 cell clusters with four conditions from 247 TF motifs. Values represent the average scaled ChromVar z-scores per clusters, illustrating dynamic regulatory transitions. **e**, Odd ratios from the enrichment between different modes of action identified in this study with established literature using different cutoffs from ChromVar variability ($> 25\%$ quantile, $>$ median and $> 75\%$ quantile, with $p.\text{adj} < 0.01$) and from auto-correlation (≥ 0.1 to 0.8). **f**, Upset plot summarizing the number of TF motifs categorized by their regulatory modes ($n = 181$ from promoter-enhancer joint ChromVar analysis). **g**, Percentage of TF motifs that are modality-specific according to the definition from **Fig. 3f** with different combinations of cutoff from ChromVar variability and correlation. **h**, Odd ratios from the enrichment between different regulatory modes identified with promoter-enhancer joint ChromVar analysis and established literature across 122 TF motifs with information from both ends.